

Handcrafted Image Retrieval: Classical and State-of-the-Art Pipelines on Oxford and Paris Buildings

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Abstract—This report presents the design, implementation, and evaluation of five handcrafted image retrieval systems on the Oxford5K and Paris6K Buildings benchmarks. The study covers both classical baselines and progressively stronger local-feature retrieval pipelines, incorporating descriptor normalization, compact aggregation, spatial verification, query expansion, and burst filtering. All systems are evaluated using mean Average Precision (mAP) on both datasets. The results show a clear performance gap between the simpler baselines and the strongest proposed pipeline, which reaches 0.7798 mAP on Oxford5K and 0.7587 mAP on Paris6K. We further provide ablation studies and qualitative analyses of representative success and failure cases.

Index Terms—image retrieval, SIFT, BoVW, Fisher vectors, VLAD, RANSAC, query expansion, Oxford Buildings, Paris Buildings

I. INTRODUCTION

The exponential growth of digital image collections in social media, surveillance systems, and multimedia archives has generated a compelling demand for efficient methods of searching and retrieving visual information. Given a query image, an image retrieval system must automatically rank a database of images according to visual similarity. This problem is particularly challenging when datasets contain images of the same landmark captured under diverse conditions of viewpoint, scale, illumination, and occlusion.

One of the most influential paradigms in classical computer vision is the Bag of Visual Words (BoVW) model [1], which adapts concepts from text retrieval to the visual domain. In this model, images are represented as histograms of quantized local descriptors, enabling efficient retrieval through inverted index structures. Building on this foundation, more expressive representations such as Fisher Vectors [2] and VLAD [2] have been proposed to capture higher-order statistics of local descriptors within a compact, fixed-length vector.

The introduction of RootSIFT [3] demonstrated that simply changing the distance metric from Euclidean to the Hellinger kernel—via an element-wise square root transformation—yields dramatic improvements across all stages of the retrieval pipeline. Subsequent work has shown that PCA combined with whitening [4] significantly reduces descriptor correlation artifacts, while geometric re-ranking via RANSAC [5] and Average Query Expansion [6] address the

precision and recall limitations of the initial ranking.

In this project, we implement and compare five image retrieval systems—two classical baselines and three stronger handcrafted retrieval pipelines—on the Oxford5K [1] and Paris6K [3] Buildings datasets. All systems are restricted to classical computer vision techniques, without the use of deep learning models.

The remainder of this report is organized as follows. Section II reviews the theoretical foundations of the implemented techniques. Section III describes the pipeline designs in detail. Section IV presents quantitative and qualitative experimental results. Section V discusses the findings, and Section VI concludes the report.

II. BACKGROUND

A. SIFT

SIFT, or Scale-Invariant Feature Transform, detects and describes distinctive local image regions. The detector builds a scale-space representation by convolving the image with Gaussian kernels at different scales:

$$L(x, y, \sigma) = I(x, y) * G(x, y, \sigma), \quad (1)$$

where I is the input image and G is a Gaussian kernel. Candidate keypoints are detected as extrema in the Difference of Gaussians:

$$D(x, y, \sigma) = L(x, y, k\sigma) - L(x, y, \sigma). \quad (2)$$

A candidate is compared against its spatial and scale neighbors. Unstable low-contrast points and edge-like responses are discarded. Each retained keypoint is then described using local gradient orientation histograms, producing a 128-dimensional descriptor.

For this retrieval task, SIFT is appropriate because landmarks contain repeatable local patterns that can be detected across different images. Since SIFT is designed to be robust to moderate changes in scale, rotation, and illumination, it provides a stronger starting point than raw pixels or global image comparison.

B. RootSIFT

Arandjelovic and Zisserman [3] observed that SIFT descriptors, being histograms, are more appropriately compared using the Hellinger kernel rather than the Euclidean distance. RootSIFT modifies SIFT descriptors through normalization and an element-wise square-root transform. Given a SIFT descriptor x , RootSIFT can be written as:

$$\tilde{x} = \sqrt{\frac{x}{\|x\|_1}}. \quad (3)$$

The descriptor is then typically L_2 normalized. Intuitively, this reduces the dominance of large histogram bins and makes descriptor comparison more robust. In the experiments, RootSIFT was evaluated as a stronger local descriptor variant, especially in combination with larger vocabularies and spatial verification.

C. HOG

Histogram of Oriented Gradients (HOG) is a global gradient-based descriptor. It represents an image by computing local gradient orientations, accumulating them into histograms over spatial cells, and normalizing these histograms over local blocks.

Given image gradients I_x and I_y , each pixel contributes to a gradient magnitude and orientation:

$$m(x, y) = \sqrt{I_x(x, y)^2 + I_y(x, y)^2},$$

$$\theta(x, y) = \arctan\left(\frac{I_y(x, y)}{I_x(x, y)}\right).$$

The final HOG descriptor is the concatenation of all normalized block histograms. HOG captures edge structure and coarse shape, but it is sensitive to global layout, viewpoint, framing, and scale. This makes it a useful baseline, but not necessarily a strong representation for landmark retrieval.

D. KAZE and AKAZE

KAZE is included as a nonlinear scale-space alternative. Instead of using a Gaussian linear scale space, KAZE constructs features through nonlinear diffusion, which aims to smooth homogeneous regions while preserving object boundaries [7]. The hypothesis behind including KAZE is that edge preservation could be beneficial for architectural imagery; this hypothesis is tested empirically in the descriptor ablation. AKAZE extends KAZE with accelerated computation using modified local difference binary descriptors.

E. Bag of Visual Words (BoVW)

A single image contains a variable number of SIFT descriptors, so direct comparison between images is inconvenient. Bag of Visual Words converts a set of local descriptors into a fixed-length vector.

First, descriptors sampled from the dataset are clustered using k -means. The resulting cluster centers form a visual vocabulary:

$$C = \{c_1, c_2, \dots, c_K\}. \quad (4)$$

Each descriptor x is assigned to its nearest visual word:

$$q(x) = \arg \min_{c_j \in C} \|x - c_j\|_2^2. \quad (5)$$

An image is then represented as a histogram of visual-word occurrences. This follows the text-retrieval analogy: local descriptors are treated as visual words, and an image is represented as a document-like word-frequency vector.

BoVW makes retrieval scalable because every image is encoded as a fixed-length histogram that can be compared with distances or similarities such as cosine similarity, chi-square distance, or histogram intersection. However, BoVW discards the spatial arrangement of descriptors. This is its main limitation: two different buildings can have similar histograms if they contain similar local patterns, even if those patterns are arranged differently.

F. Fisher Vectors

The Fisher Vector (FV) [2] encodes an image by measuring how its local descriptors deviate from the learned GMM parameters. In this work, Fisher Vectors are computed with `scikit-image`'s `fisher_vector` function using `improved=True`.

For a GMM with K components and descriptors of dimension D , the Fisher Vector dimension is

$$\dim(FV) = K + 2KD.$$

The improved Fisher Vector applies power normalization and ℓ_2 normalization:

$$z_i \leftarrow \text{sign}(z_i)|z_i|^{0.5}, \quad z \leftarrow \frac{z}{\|z\|_2}.$$

These steps reduce the effect of large components and make cosine similarity more stable.

G. VLAD

The Vector of Locally Aggregated Descriptors (VLAD) [2] is a compact image representation derived from a visual vocabulary $\mathcal{C} = \{c_1, \dots, c_K\}$ learned by k -means. For each visual word c_i , VLAD accumulates the residuals of all descriptors assigned to it:

$$V_{i,j} = \sum_{x: \text{NN}(x)=c_i} (x_j - c_{i,j}), \quad (6)$$

where j indexes the descriptor dimension. The full VLAD vector is the concatenation of all per-word residuals, followed by L_2 normalization. Compared to Bag-of-Visual-Words, VLAD retains richer information about the distribution of descriptors within each Voronoi cell.

H. PCA Whitening

Jégou and Chum [4] showed that applying PCA with whitening to VLAD vectors improves retrieval by two complementary effects. Mean subtraction addresses *negative evidence*: visual words absent from both the query and a database image become informative under a shifted similarity measure. Eigenvalue scaling (whitening) de-correlates dimensions and reduces the effect of co-occurring visual words, which would otherwise dominate the similarity score. Each projected vector is L2-renormalized after whitening, which is critical for correct cosine similarity computation.

The whitening step decorrelates the projected dimensions by scaling each component by the inverse square root of its eigenvalue:

$$\hat{X} = \frac{\text{diag}(\lambda_1^{-1/2}, \dots, \lambda_{D'}^{-1/2}) P^\top X}{\|\text{diag}(\lambda_1^{-1/2}, \dots, \lambda_{D'}^{-1/2}) P^\top X\|} \quad (7)$$

This corrects co-occurrence artifacts caused by correlated visual words and reduces the dimensionality of the representation to D' components, typically $D' = 128$ [4].

I. Visual Vocabulary and ASMK Encoding

A visual vocabulary is learned by k-means from a large sample of local descriptors. Let

$$\mathcal{C} = \{c_1, c_2, \dots, c_K\} \quad (8)$$

be the set of visual words. Each descriptor x is assigned to its nearest centroid:

$$q(x) = \arg \min_{c_j \in \mathcal{C}} \|x - c_j\|_2^2. \quad (9)$$

A Bag of Visual Words representation would only count visual-word occurrences. ASMK instead preserves residual information inside each visual-word cell. For each descriptor assigned to a visual word c , the normalized residual is computed as

$$\phi(x) = \frac{x - c}{\|x - c\|_2}. \quad (10)$$

For each visual word, descriptors assigned to that cell are aggregated:

$$V_c(\mathcal{X}) = \sum_{x:q(x)=c} \phi(x), \quad (11)$$

and the aggregated vector is normalized:

$$\hat{V}_c(\mathcal{X}) = \frac{V_c(\mathcal{X})}{\|V_c(\mathcal{X})\|_2}. \quad (12)$$

The ASMK similarity between two images $\mathcal{X}$ and $\mathcal{Y}$ is computed over shared visual words:

$$K(\mathcal{X}, \mathcal{Y}) = \gamma(\mathcal{X})\gamma(\mathcal{Y}) \sum_{c \in \mathcal{C}} w_c \sigma_\alpha \left(\hat{V}_c(\mathcal{X})^\top \hat{V}_c(\mathcal{Y}) \right), \quad (13)$$

where w_c is an IDF-like weight, γ is a global normalization factor, and σ_α is a selectivity function that suppresses weak matches [8]. The central assumption is that two views of the same landmark should share discriminative visual words and have compatible residual directions within those words.

J. Normalization, Similarity Metrics, and TF-IDF

Histogram normalization is important because images can produce different numbers of keypoints. Without normalization, highly textured images may obtain large histogram counts simply because they contain more descriptors. The experiments tested L_1 , L_2 , and square-root based normalization.

The initial experiments compared cosine similarity, chi-square distance, and histogram intersection. Cosine similarity compares the direction of two vectors:

$$\cos(q, d) = \frac{q^\top d}{\|q\|_2 \|d\|_2}. \quad (14)$$

Chi-square distance is commonly used for histograms because it emphasizes relative bin differences:

$$\chi^2(q, d) = \frac{1}{2} \sum_i \frac{(q_i - d_i)^2}{q_i + d_i + \epsilon}. \quad (15)$$

TF-IDF weighting was also tested. The idea is to down-weight visual words that appear in many images and upweight more discriminative words. If h_{ij} is the count of visual word j in image i , a simplified TF-IDF weighting is:

$$\text{tfidf}_{ij} = h_{ij} \log \frac{N}{n_j}, \quad (16)$$

where N is the number of images and n_j is the number of images containing visual word j .

K. RANSAC Spatial Verification

Spatial verification via RANSAC [5] re-ranks the top- k retrieved images by estimating a geometric transformation between the query and each candidate. Each keypoint correspondence hypothesizes a transformation; the hypothesis with the greatest number of inliers is selected. Philbin et al. [5] showed that spatial re-ranking consistently and significantly improves mAP over the bag-of-words baseline across all vocabulary sizes, demonstrating its importance as a precision-boosting stage.

First, SIFT descriptors from the query bounding box are matched against descriptors from a full database image. For each query descriptor, the two nearest candidate descriptors are retrieved. Lowe's ratio test accepts a match only when the nearest neighbor is sufficiently better than the second nearest neighbor:

$$\frac{d_1}{d_2} < \tau, \quad (17)$$

where d_1 and d_2 are the nearest and second-nearest descriptor distances. This removes ambiguous matches, especially those caused by repeated windows or similar facade patterns.

After Lowe filtering, RANSAC estimates a homography H between the query region and the candidate image. A match (p_i, p'_i) is considered an inlier if:

$$\|p'_i - Hp_i\|_2 < \epsilon. \quad (18)$$

The candidate image is then scored by the number of RANSAC inliers. The intuition is that a true landmark match should support a single plausible projective transformation,

while false positives may share visual words but fail geometric consistency.

L. Average Query Expansion (AQE)

Query expansion refines the query representation using the top-ranked retrieved images. If q is the original query vector and $d_1, \dots, d_M$ are the top M database vectors, the expanded query is:

$$q' = \alpha q + (1 - \alpha) \frac{1}{M} \sum_{i=1}^M d_i. \quad (19)$$

The expanded query is normalized and used to rerank the database.

The method is based on the assumption that the first retrieved images are likely to be relevant. If this assumption holds, query expansion improves recall by adding complementary visual evidence from other views of the same landmark. If the top results contain false positives, however, the query can drift toward the wrong landmark. This risk is central to the interpretation of the results: query expansion helped in some stages, but it was not always reliable before spatial verification.

Average Query Expansion [6] enriches the query representation by averaging the feature vectors of the top spatially verified results with the original query vector. The expanded query is then used to re-issue the search against the database. This post-processing step improves recall by incorporating visual evidence from confirmed true positives into the query [3].

III. METHODOLOGY

Dataset and Evaluation Protocol

We evaluate all systems on the Oxford5K dataset (5,062 images, 55 queries over 11 landmarks) and the Paris6K dataset (6,392 images, 55 queries over 11 landmarks) [3]. Both datasets are evaluated independently; for each dataset, the retrieval system is trained and indexed on that dataset's images. Performance is measured by mean Average Precision (mAP) as defined by the ground-truth annotations provided with each dataset.

Query descriptors are extracted from the annotated bounding box region of each query image. Database descriptors are extracted from the full image. Evaluation follows the standard protocol [1]: images are ranked by descending similarity score and AP is computed per query before averaging.

The main metric is mean Average Precision (mAP). Precision at rank k is also reported to measure the fraction of relevant images among the top- k retrieved results:

$$\text{Precision@}k = \frac{\# \text{ relevant images in the top } k}{k} \quad (20)$$

For one query, Average Precision is:

$$AP = \frac{1}{N} \sum_{k=1}^M P(k) \text{rel}(k), \quad (21)$$

where $P(k)$ is precision at rank k , $\text{rel}(k)$ indicates whether the item at rank k is relevant, M is the ranking length, and N is the total number of relevant images for that query.

The final mAP is:

$$\text{mAP} = \frac{1}{Q} \sum_{i=1}^Q AP_i, \quad (22)$$

where Q is the number of queries.

1) *System B1: SIFT + BoVW*: The final SIFT-centered pipeline is

- 1) Extract SIFT or multi-scale SIFT descriptors.
- 2) For query images, keep only descriptors inside the query bounding box.
- 3) Train a BoVW vocabulary using k -means.
- 4) Encode each image as a visual-word histogram.
- 5) Normalize descriptors and rank database images with cosine similarity or another tested metric.
- 6) Apply spatial verification to the top N candidates using Lowe ratio filtering and RANSAC homography estimation.
- 7) Apply verified query expansion to the spatially verified ranking.
- 8) Apply a second spatial verification pass to reduce query drift after expansion.

The best final configuration used multi-scale SIFT with vocabulary size $K = 4096$, L_2 normalization, cosine similarity, spatial verification over the top 150 candidates, Lowe ratio $\tau = 0.65$, verified query expansion with top 3 results and $\alpha = 0.5$, followed by a second spatial verification pass.

A. System 2: HOG + Fisher Vectors

This pipeline uses HOG as a global image descriptor combined with the Fisher Vector encoding scheme. A compact diagram of the workflow is provided in Appendix B, Fig. 17.

- 1) **Feature extraction**: HOG descriptors are computed over a dense grid on the image. Query features are extracted from the bounding box crop; database features from the full image.
- 2) **GMM training**: A Gaussian Mixture Model (GMM) with k components is trained on a subset of HOG descriptors. The GMM captures the generative model of the descriptor distribution.
- 3) **Fisher Vector encoding**: Each image is encoded as a Fisher Vector by computing the gradient of the log-likelihood of its HOG descriptors with respect to the GMM parameters. This yields a $(2d + 1) \times k - 1$ dimensional vector, where d is the descriptor dimension.
- 4) **Retrieval**: Images are ranked by cosine similarity between the query Fisher Vector and all database vectors.

We compare three progressively stronger classical descriptor strategies:

$$\begin{aligned} \text{HOG baseline} &\rightarrow \text{RootSIFT} + \text{Fisher Vector} \\ &\rightarrow \text{HOG-Fisher fusion.} \end{aligned} \quad (23)$$

HOG serves as a global gradient baseline, Fisher Vector encodes local RootSIFT descriptors through a GMM-based aggregation scheme, and the fusion experiment evaluates whether global gradient structure adds complementary information to the Fisher representation.

a) HOG baseline: HOG represents an image through local histograms of gradient orientations accumulated on a regular grid. It is a useful baseline because it captures edge structure and coarse shape, but for landmark retrieval it remains sensitive to viewpoint, framing, and spatial misalignment. The descriptor-extraction flow is integrated into the full B1 pipeline shown in Appendix B, Fig. 17.

b) RootSIFT + Fisher Vector: The Fisher pipeline replaces the global HOG grid with local keypoint descriptors. SIFT descriptors are first transformed into RootSIFT by L1 normalization followed by an element-wise square root [3]. The local descriptors are then projected from 128 to 64 dimensions using PCA and encoded with an improved Fisher Vector built from a K -component GMM. For a descriptor dimension D and GMM size K , the Fisher representation has dimension $K + 2KD$; for the best configuration with $K = 64$ and $D = 64$, this yields 8256 dimensions. The corresponding descriptor pipeline is summarized in Appendix B, Fig. 19.

c) HOG-Fisher fusion: To test whether global structure complements local aggregation, we normalize the HOG and Fisher branches independently, concatenate them with scalar weights, and normalize the final vector once more:

$$v = \text{concat}(w_F L2(FV), w_H L2(HOG)), \quad v \leftarrow L2(v). \quad (24)$$

We fix $w_F = 1.0$ and evaluate $w_H \in \{0.10, 0.25, 0.50\}$. The late-fusion scheme is shown in Appendix B, Fig. 20.

d) Retrieval protocol: After descriptor extraction, each query descriptor is compared with all database descriptors using cosine similarity. Since all descriptors are L_2 -normalized, cosine similarity is equivalent to a dot product. AP is computed using the official Oxford evaluation script and mAP is obtained by averaging AP over the 55 queries. The simplified retrieval-and-evaluation protocol is diagrammed in Appendix B, Fig. 21.

B. System 3: RootSIFT + VLAD + PCA + Whitening + RANSAC + AQE

This system follows a two-stage retrieval design with offline indexing and online query-time reranking. The full processing chain appears in Appendix B, Fig. 22.

1) Offline Stage: The offline stage processes all database images once and stores the resulting index.

- 1) **RootSIFT extraction:** Each image is resized so that its longest side is at most 1024 pixels. OpenCV SIFT is applied without a keypoint limit, detecting approximately 3 300 keypoints per image. Descriptors are transformed to RootSIFT using Eq. (3).
- 2) **Visual vocabulary:** A vocabulary of $K = 256$ visual words is learned by MiniBatch k-means on a random

sample of 2 000 images, using up to 150 descriptors per image.

- 3) **VLAD encoding:** Each image is encoded as a VLAD vector of dimension $K \times 128 = 32\,768$ using Eq. (6), with signed square-root power normalization [2] applied before the final L2 normalization to reduce the effect of bursty visual elements.
- 4) **PCA whitening:** PCA with whitening is fitted on the VLAD database matrix and reduces the representation to $D = 512$ dimensions. Each projected vector is L2-renormalized.
- 5) **SIFT cache for RANSAC:** To avoid re-extracting SIFT at query time, the top-500 keypoints (by SIFT response) are saved per image in compressed .npz files. Loading a cached file takes approximately 2 ms versus 400 ms for live extraction, reducing the RANSAC stage from ~ 20 s to ~ 0.1 s per query.

2) Online Stage: The online stage processes each query at retrieval time.

- 1) **Query encoding:** The query image, restricted to the annotated bounding box, is encoded with the same RootSIFT $\rightarrow$ VLAD $\rightarrow$ PCA whitening pipeline, producing a 512-dimensional L2-normalized vector.
- 2) **Initial retrieval:** Cosine similarity is computed between the query vector and all database vectors via a single dot product:

$$\text{sim}(q, v_i) = q^\top v_i, \quad (25)$$

since all vectors are L2-normalized. Results are ranked in descending order of similarity.

- 3) **RANSAC spatial re-ranking:** The top-50 candidates are re-ranked by geometric verification. Keypoint descriptors are loaded from the SIFT cache. FLANN matching with Lowe's ratio test (threshold 0.75) selects correspondences, and RANSAC estimates a homography with a reprojection threshold of 10 pixels. Candidates with at least 6 inliers are promoted above unverified images, sorted by inlier count.
- 4) **Average Query Expansion:** The query vector is averaged with the top-10 post-RANSAC database vectors, and a second retrieval pass is performed with the expanded query.

C. System 4: AKAZE/KAZE + K-Means + ASMK + RANSAC + AQE

This pipeline replaces the Gaussian-based SIFT detector with the nonlinear-scale-space KAZE/AKAZE detector [7], and uses the ASMK encoding for improved selectivity and burstiness handling. The full processing chain appears in Appendix B, Fig. 23.

- 1) **KAZE/AKAZE extraction:** Features are detected as maxima of the scale-normalized Hessian determinant in the nonlinear scale space. Nonlinear diffusion using

AOS techniques preserves object boundaries while reducing noise [7]. Each keypoint is described by a 128-dimensional descriptor.

- 2) **Vocabulary training:** K-Means is applied to a training set of KAZE/AKAZE descriptors.
- 3) **ASMK encoding:** Descriptors are encoded using ASMK: residuals per visual word are aggregated and L2-normalized, and a selectivity function σ_α suppresses weak correspondences and handles bursts [8]. As an alternative, VLAD encoding is also evaluated.
- 4) **Initial ranking:** Images are ranked by the ASMK similarity score using an inverted file structure.
- 5) **RANSAC spatial verification:** The top-150 results are re-ranked by geometric inlier count estimated by RANSAC [5].
- 6) **AQE:** Verified top results are averaged with the query representation and a new query is issued [6].

D. System 5: RootSIFT + Burst Filtering + ASMK + RANSAC

This pipeline combines Hessian-Affine feature detection, RootSIFT descriptors, early burst filtering through connected components, and image indexing with the Aggregated Selective Match Kernel (ASMK) using an inverted file structure [3], [8], [9].

- 1) **Feature extraction:** Hessian-Affine [10] detects scale-invariant interest points with the Hessian operator and iteratively adapts each region into an ellipse to improve affine invariance under strong viewpoint changes. To reduce rotational ambiguity, the detector uses the gravity-vector assumption, constraining the ellipse transform to a lower-triangular form and keeping the vertical axis fixed. RootSIFT descriptors are then extracted on the detected regions to provide a robust local appearance representation.
- 2) **Visual vocabulary:** A flat k -means vocabulary is learned on an independent descriptor set. Instead of storing only visual-word counts as in classical BoVW, the method keeps residual vectors, which preserve the exact difference between each local descriptor and its assigned centroid. Very large vocabularies, for example around 65k visual words, are preferred because they increase sparsity and make the inverted-file lookup faster at query time.
- 3) **Image representation:** Before assignment, an early burst-filtering stage builds a similarity graph and runs connected components to group highly similar local descriptors inside the same image [9]. Each group is averaged into a meta-feature, reducing the number of descriptors to roughly 80–85% of the original volume. This reduction is applied asymmetrically: only database images are filtered, while the query image remains unchanged to avoid losing valid matches. If several meta-features fall into the same visual word, their residual vectors are summed and L2-normalized, producing one aggregated residual per active visual word.

- 4) **Similarity and retrieval:** Similarity is computed with ASMK [8]. For each shared visual word, the kernel evaluates the dot product between the aggregated residuals of the query and database image, then applies a polynomial selectivity function with threshold σ_α to suppress weak matches. The final score is scaled with a global normalization factor γ so that an image compared with itself obtains score 1. Retrieval is accelerated with an inverted file, which skips visual words absent from the query and lets missing words contribute zero to the final score.

This pipeline closely follows the retrieval framework of Philbin et al. [1], with the RootSIFT improvement of Arandjelović and Zisserman [3]. Each image is represented as a sparse tf-idf weighted Bag-of-Visual-Words vector, initial retrieval is performed with cosine similarity, and the top-ranked results are reordered by geometric verification using a homography estimated with RANSAC.

- 1) **Feature extraction:** OpenCV SIFT is used with a reduced contrast threshold to increase the number of points in low-contrast facade regions. Each image is resized so that its longest side does not exceed 1024 pixels, while preserving the scale factor needed to reproject the query bounding box correctly. A 128-dimensional SIFT descriptor is computed for each keypoint and converted to RootSIFT by L1 normalization followed by an element-wise square root, which is equivalent to comparing the original SIFT descriptors under the Hellinger kernel.
- 2) **Visual vocabulary:** A visual vocabulary is trained with *MiniBatchKMeans* on a stratified sample of RootSIFT descriptors from the Oxford collection, using up to 300 descriptors per image to avoid overrepresenting images with many keypoints. A vocabulary of $K = 20,000$ visual words is used as the reference configuration, balancing discriminative power with the available memory and training budget. Each database descriptor is hard-assigned to its nearest visual word under Euclidean distance.
- 3) **Image representation:** Each image is encoded as a sparse BoVW histogram of dimension K , counting the occurrences of assigned visual words. The histogram is then weighted with the standard tf-idf scheme,

$$\text{idf}(w) = \log\left(\frac{N+1}{\text{df}(w)+1}\right) + 1, \quad (26)$$

where N is the number of database images and $\text{df}(w)$ is the number of images containing visual word w . The resulting vector is L2-normalized so cosine similarity reduces to a dot product. For the query, RootSIFT descriptors are extracted on the full image, then filtered to keep only keypoints whose coordinates fall inside the resized benchmark bounding box, ensuring the query vector represents only the selected object.

- 4) **Similarity and retrieval:** Initial retrieval is performed by cosine similarity between the query vector and the

sparse tf-idf matrix of the database. The top-200 results are then spatially reranked. Candidate correspondences are formed by matching query and database keypoints that share the same visual word, and a homography is estimated with RANSAC using an 8-pixel reprojection threshold. Verified images with at least 4 inliers are promoted to the top of the ranking and sorted by inlier count, while non-verified images keep their original tf-idf order. The initial search is accelerated with a CSR sparse matrix, and the spatial reranking cost remains linear in the top- N shortlist rather than the full database size.

IV. EXPERIMENTS AND RESULTS

A. System 1

The experimental process was organized into six rounds:

- **Round 1:** SIFT BoVW baseline. Tested vocabulary sizes $K \in \{128, 256, 512\}$, multiple normalizations, cosine, chi-square, histogram intersection, and TF-IDF.
- **Round 2:** Larger BoVW search and query expansion. Tested vocabulary sizes up to $K = 1024$ and query expansion with $M \in \{3, 5, 10\}$ and $\alpha \in \{0.25, 0.5, 0.75\}$.
- **Round 3:** Global descriptor fusion and first spatial verification. Tested HOG, HSV, score-level fusion, and spatial verification with top 50 or 100 candidates.
- **Round 4:** Spatial verification tuning. Increased reranking depth to top 150 and tested stricter Lowe ratios around 0.65.
- **Round 5:** Stronger local evidence. Tested larger vocabularies ($K = 2048, 4096$), SIFT versus RootSIFT, verified query expansion, and a second spatial verification pass.
- **Round 6:** Multi-scale SIFT. Extracted SIFT descriptors at scales 0.75, 1.0, and 1.25, then reused the strongest BoVW, spatial verification, verified QE, and SV2 pipeline.

Table I summarizes the best result obtained at each round. The progression shows consistent improvement from the initial BoVW baseline to the final multi-scale SIFT pipeline.

TABLE I
BEST MAP PER EXPERIMENTAL ROUND. PARIS WAS FIRST VALIDATED FROM ROUND 2 ONWARD.

Round	Main method	Oxford	Paris
R1	SIFT BoVW baseline	0.1952	–
R2	Larger vocabulary + QE	0.2184	0.3724
R3	First spatial verification	0.2742	0.3922
R4	Tuned spatial verification	0.2991	0.4062
R5	Larger vocabulary / RootSIFT / verified QE / SV2	0.3548	0.4311
R6	Multi-scale SIFT + SV + verified QE + SV2	0.3634	0.4454

1) *Round 1: SIFT BoVW Baseline:* The first round established a SIFT BoVW baseline. The strongest Oxford configuration was `sift_bovw_tfidf_k256_l2_chisquare`,

with mAP 0.1952. The corresponding plain BoVW configuration reached 0.1950, so TF-IDF improved the best result only slightly.

This result suggests that BoVW already provides a meaningful retrieval signal, but that simple histogram weighting is not enough to solve the main ambiguity. The early chi-square results were stronger than cosine, indicating that relative histogram-bin differences were useful in the initial small-vocabulary setting.

2) *Round 2: Larger Vocabulary and Query Expansion:* Round 2 expanded the vocabulary size up to $K = 1024$ and tested query expansion. The best Oxford result improved to 0.2184 with `sift_bovw_k1024_l2_cosine_qe_top3_alpha0p25`. The same configuration reached 0.3724 mAP on Paris.

This changed the best setup from $K = 256$ with chi-square to $K = 1024$ with L_2 normalization, cosine similarity, and query expansion. The best query expansion used only the top 3 images, while larger expansion sets were weaker. This supports the interpretation that query expansion helps when the top results are reliable, but becomes risky when too many potentially incorrect results are included.

3) *Round 3: Fusion Attempt and First Spatial Verification:* Round 3 compared global HOG/HSV fusion against spatial verification. The best HOG fusion on Oxford improved only from 0.2184 to 0.2200, and HSV/global fusion variants were similarly marginal. On Paris, the best global fusion improved from 0.3724 to 0.3769. These gains were small relative to the cost and complexity of adding additional global features.

By contrast, spatial verification substantially improved retrieval. The best Round 3 configuration reached 0.2742 on Oxford and 0.3922 on Paris. This was the first major jump in the project.

The result confirms that the main limitation of BoVW is not only descriptor quality, but also the lack of spatial layout. HOG and HSV provide global information, but they do not verify whether local patches align consistently. Spatial verification directly addresses this weakness.

4) *Round 4: Spatial Verification Tuning:* Round 4 tuned the spatial verification stage. Increasing the reranking depth from top 100 to top 150 and using a stricter Lowe ratio of 0.65 improved Oxford to 0.2991. The best Oxford result did not use query expansion before spatial verification:

`sift_bovw_k1024_l2_cosine_spatial_verify_top150_ratio0p65_inliers`.

This suggests that query expansion is not always beneficial before geometric reranking. Query expansion can improve the first-stage representation, but it can also move the query toward false positives. The best Paris result still used query expansion before spatial verification, so the effect is dataset-dependent.

5) *Round 5: Larger Local Vocabulary, RootSIFT, Verified QE, and SV2:* Round 5 shifted the focus from tuning only the geometric reranker to improving the local vocabulary that feeds it. SIFT and RootSIFT were tested with vocabulary sizes $K = 2048$ and $K = 4096$. Verified query expansion

was applied after spatial verification, and a second spatial verification pass was then used to clean the expanded ranking.

The best Oxford result improved to 0.3548. The best Paris result improved to 0.4311. Increasing vocabulary size was the most consistently helpful factor. RootSIFT was useful but dataset-dependent: it became strongest on Paris, but plain SIFT remained slightly better on Oxford under some configurations.

Verified query expansion alone did not consistently beat the previous best, but verified QE followed by a second spatial verification pass was strong. This indicates that query expansion can introduce useful new candidates, but even after verification it may still introduce drift. The second spatial verification pass suppresses this drift by reapplying geometric consistency.

6) *Round 6: Multi-scale SIFT*: Round 6 tested multi-scale SIFT with scales 0.75, 1.0, and 1.25. The strongest configuration used $K = 4096$, spatial verification with top 150 and Lowe ratio 0.65, verified QE with top 3 and $\alpha = 0.5$, and a second spatial verification pass. It reached 0.3634 mAP on Oxford and 0.4454 mAP on Paris.

TABLE II
TOP ROUND 6 RESULTS.

Configuration	Oxford mAP
Multi-scale SIFT $K = 4096$ + SV + verified QE + SV2	0.3634
Multi-scale SIFT $K = 2048$ + SV + verified QE + SV2	0.3485
Multi-scale SIFT $K = 4096$ + SV	0.3355
Multi-scale SIFT $K = 4096$ + SV + verified QE	0.3332
Multi-scale SIFT $K = 2048$ + SV	0.3194
Multi-scale SIFT $K = 2048$ only	0.2407
Multi-scale SIFT $K = 4096$ only	0.2345

The gain over Round 5 was smaller than the gain introduced by spatial verification, but it was consistent across both datasets. This matches the nature of the method: multi-scale extraction adds more local evidence, but it does not fundamentally change the BoVW representation. The final result is therefore best understood as an incremental improvement over a strong SIFT-centered retrieval pipeline.

B. System 2

a) *HOG baseline*: Table III reports the HOG-only configurations. The best HOG-only configuration, 256-c16-o12, reaches mAP 0.0350, confirming that a rigid global gradient descriptor is too weak for Oxford landmark retrieval when viewpoint and crop variation are substantial.

TABLE III
OXFORD5K HOG-ONLY RETRIEVAL RESULTS FOR THE
DESCRIPTOR-LEVEL STUDY.

Experiment	Img.	Ori.	Cell	Block	mAP	P@1	P@5	P@10
256-c16-o12	256 ²	12	(16, 16)	(2, 2)	0.0350	0.2364	0.1309	0.1127
256-c16-o9	256 ²	9	(16, 16)	(2, 2)	0.0320	0.2364	0.1236	0.1055
384-c16-o9	384 ²	9	(16, 16)	(2, 2)	0.0250	0.1636	0.0945	0.0891
256-c8-o9	256 ²	9	(8, 8)	(2, 2)	0.0223	0.1273	0.0836	0.0709

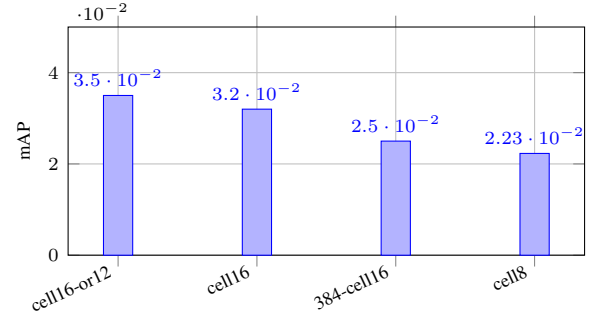


Fig. 1. mAP comparison among HOG-only configurations. All HOG configurations remain below 0.04 mAP.

b) *RootSIFT + Fisher Vector*: Table IV reports the Fisher Vector configurations. Fisher Vector substantially improves over HOG. Increasing the GMM size from $K = 32$ to $K = 64$ raises mAP from 0.3982 to 0.4237, showing that a richer local descriptor distribution yields a more discriminative global encoding.

TABLE IV
OXFORD5K FISHER VECTOR RETRIEVAL RESULTS

Experiment	K	PCA	mAP	P@1	P@5
fisher-k64-pca64	64	64	0.4237	1.0000	0.7491
fisher_k32_pca64	32	64	0.3982	0.9818	0.7091

c) *HOG-Fisher fusion*: Table V reports the late-fusion results. Adding HOG does not improve the Fisher-only baseline: a very small HOG contribution preserves performance, but larger HOG weights consistently degrade mAP.

TABLE V
OXFORD5K HOG-FISHER FUSION RESULTS

Experiment	Mode	K	PCA	HOG w.	mAP	P@1	P@5
f-k64-p64	Fisher	64	64	0.00	0.4237	1.0000	0.7491
hf-hw010	HOG + Fisher	64	64	0.10	0.4236	1.0000	0.7455
hf-hw025	HOG + Fisher	64	64	0.25	0.4207	1.0000	0.7382
hf-hw050	HOG + Fisher	64	64	0.50	0.4077	0.9818	0.7200

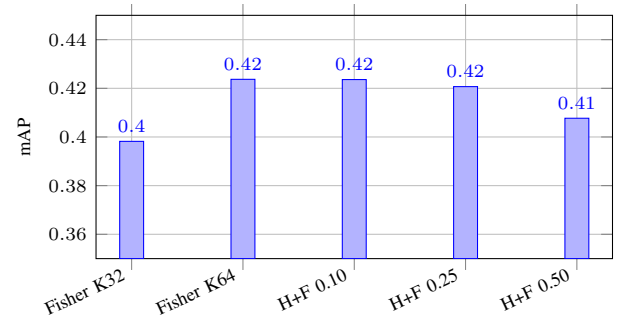


Fig. 2. mAP comparison across Fisher and HOG-Fisher configurations. Fisher with $K = 64$ and PCA64 obtained the best score.

C. System 3

Table VI reports the contribution of each component in the pipeline

TABLE VI
ABLATION STUDY FOR MAP

Configuration	Oxford5k	Paris6k
Baseline (VLAD+PCA-Wh)	0.3549	0.4310
Baseline + RANSAC	0.3285	0.4320
Baseline + AQE	0.4591	0.5955
Baseline + RANSAC + AQE	0.4726	0.6168

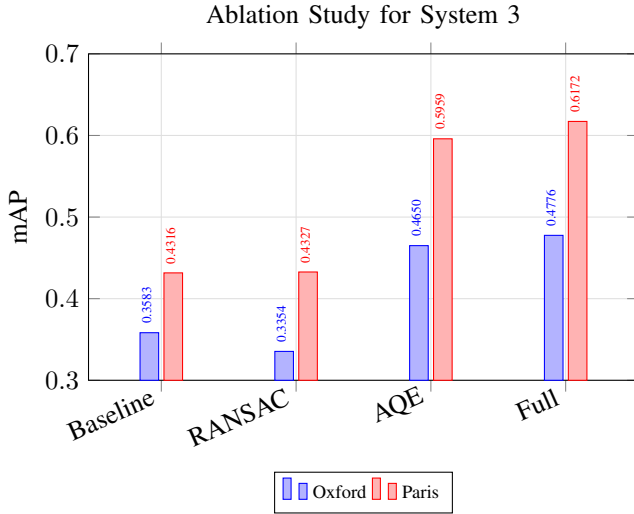


Fig. 3. mAP comparison for the ablation study on Oxford and Paris. Values above each bar report the exact mAP.

The ablation clearly shows that spatial verification is the single most impactful component, producing a mAP gain of approximately +0.10 over the best BoVW baseline. This is consistent with the findings of Philbin et al. [5], who showed that spatial re-ranking consistently improves over the BoVW baseline regardless of vocabulary size. Verified AQE provides further improvement, confirming the effectiveness of query expansion when the initial ranking contains a sufficient number of reliable true positives [6].

The transition from SIFT to RootSIFT provides a consistent improvement of approximately 0.02–0.03 mAP, in line with the results of Arandjelovic and Zisserman [3]. Larger vocabularies ($k = 4096$) further improve first-stage ranking quality.

D. System 4

The main configuration is a medium-strength preset designed to improve over the preliminary ablation while keeping memory usage feasible. ASMK is run with `binary=False`, meaning that aggregated residuals are stored as `float32` vectors. This increases memory consumption compared with binary ASMK but avoids relying on the compiled

`asmk.hamming` extension, which was not compatible with the execution environment.

TABLE VII
MAIN ASMK CONFIGURATION USED FOR OXFORD AND PARIS.

Parameter	Value
Descriptor	SIFT
Maximum image side	900
Maximum features/image	3000
Visual words	16384
Training descriptors	700000
ASMK top- k	None
SV top- k	80
QE top- k	5
Lowe ratio	0.75
RANSAC threshold	5.0
ASMK binary mode	False

The image side is capped at 900 pixels to reduce SIFT extraction cost while preserving facade-level structure. The number of local features is limited to 3000 per image to avoid allowing highly textured images to dominate the descriptor table. A vocabulary of 16384 visual words is used as a compromise between discrimination and memory usage. The training sample of 700000 descriptors provides sufficient visual diversity without using all descriptors in the datasets. Spatial verification is restricted to the top 80 candidates because RANSAC is expensive and is most useful where positives are likely to occur. Query expansion uses the top 5 results to reduce the probability of query drift.

1) *Descriptor Ablation*: The first experiment compares SIFT, RootSIFT, and KAZE under the same ASMK retrieval framework. The purpose is to select a descriptor for the stronger main run.

TABLE VIII
PRELIMINARY DESCRIPTOR ABLATION UNDER THE SAME ASMK FRAMEWORK.

Descriptor	ASMK	+SV	+QE	+SV+QE
SIFT	0.241668	0.303278	0.296009	0.379705
RootSIFT	0.240512	0.297906	0.283046	0.359126
KAZE	0.128909	0.174892	0.159155	0.227531

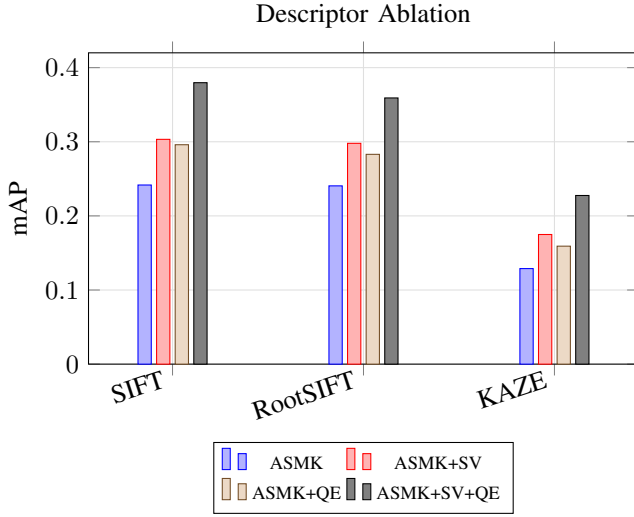


Fig. 4. Descriptor-level comparison under the same retrieval framework. SIFT obtains the highest mAP in all evaluated variants.

SIFT achieves the highest mAP in every variant. RootSIFT remains close to SIFT but does not improve it in this ASMK setting. KAZE performs substantially worse, indicating that its nonlinear scale-space features were less suitable for the implemented vocabulary and residual-aggregation scheme. Based on this result, SIFT is used as the main descriptor in the final run.

2) *Main Retrieval Results*: Table X reports the final results for SIFT + ASMK on Oxford and Paris.

TABLE IX
MAIN SIFT + ASMK RESULTS ON OXFORD AND PARIS.

Dataset	ASMK	+SV	+QE	+SV+QE
Oxford	0.632691	0.612014	0.668028	0.643018
Paris	0.695318	0.691916	0.743108	0.748987

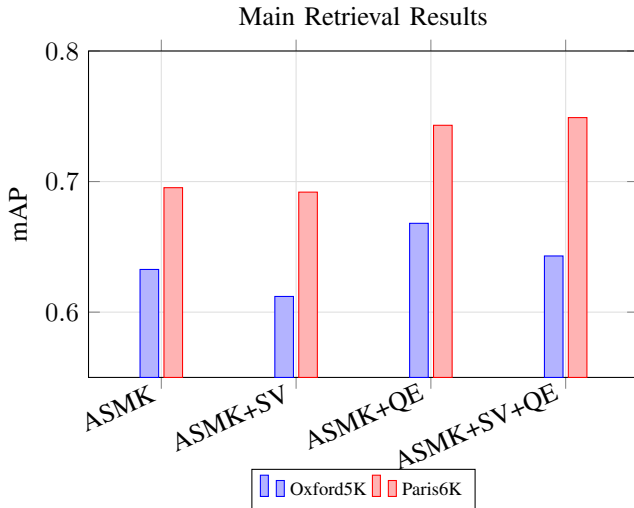


Fig. 5. Main mAP comparison for SIFT + ASMK. Query expansion is beneficial on both datasets, while spatial verification behaves differently across datasets.

On Oxford, the best configuration is ASMK + QE, reaching 0.668028 mAP. Spatial verification decreases performance both when applied alone and when combined with query expansion. On Paris, the best configuration is ASMK + SV + QE, reaching 0.748987 mAP, although ASMK + QE is very close at 0.743108 mAP.

E. Effect of Post-processing Components

To isolate the contribution of spatial verification and query expansion, Fig. 6 reports the mAP change relative to the ASMK baseline.

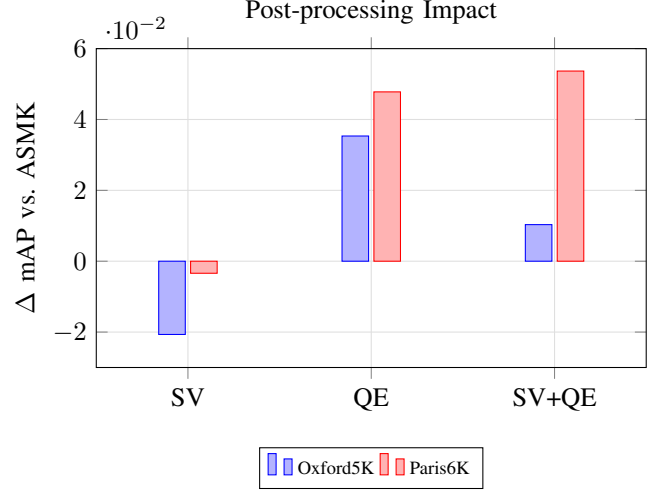


Fig. 6. Incremental effect of spatial verification and query expansion relative to ASMK. Query expansion is the only consistently positive component across datasets.

Query expansion improves mAP on both datasets. The gain is +0.035337 on Oxford and +0.047790 on Paris. Spatial verification alone decreases mAP on both datasets. This suggests that the ASMK initial ranking is already strong and that the additional geometric signal may be too restrictive or noisy under the selected parameters. Nevertheless, the combination SV+QE is beneficial on Paris, where it provides the best final result.

F. System 5

System 5 introduces Burst Filtering and transitions to Root-SIFT within the ASMK framework. This configuration yields a remarkably strong initial baseline compared to previous systems. On Oxford, the ASMK baseline achieves 0.7701 mAP (up from 0.6326 in System 4), and on Paris, it reaches 0.7267 mAP (up from 0.6953).

On Oxford, the best overall configuration is ASMK + SV, peaking at 0.7798 mAP. Unlike the behavior observed in System 4, the addition of Query Expansion (QE) degrades performance on this dataset. On Paris, the highest performance is achieved by the combined ASMK + SV + QE pipeline, reaching 0.7587 mAP.

TABLE X
MAIN BURST FILTERING + ROOTSIFT + ASMK RESULTS ON OXFORD
AND PARIS.

Dataset	ASMK	+SV	+QE	+SV+QE
Oxford	0.7701	0.7798	0.7313	0.7684
Paris	0.7267	0.7208	0.6584	0.7587

To analyze the specific impact of spatial verification and query expansion in this upgraded pipeline, Figure 7 illustrates the mAP change relative to the Burst Filtering + RootSIFT + ASMK baseline.

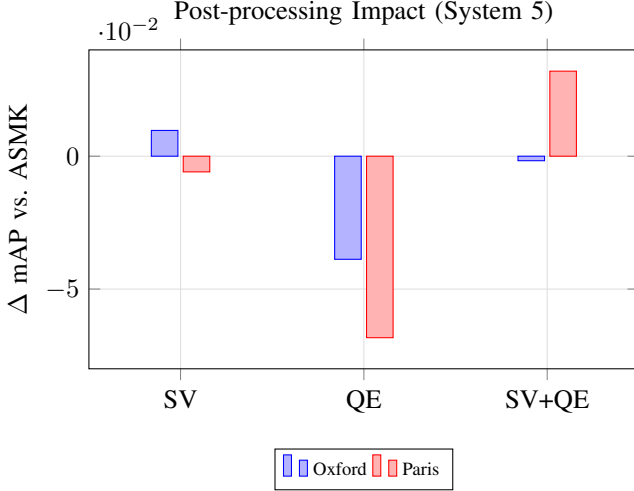


Fig. 7. Incremental effect of spatial verification and query expansion relative to the System 5 ASMK baseline. QE alone is detrimental, but the SV+QE combination shows strong synergy on Paris.

The behavior of post-processing components in System 5 sharply contrasts with System 4. Because Burst Filtering and RootSIFT already produce a highly purified and accurate initial ranking, standard unverified Query Expansion introduces severe query drift. This is evident in the sharp mAP drops when QE is applied alone (-0.0388 on Oxford and -0.0683 on Paris). Spatial Verification (SV) alone provides a slight improvement on Oxford ($+0.0097$) but a minor decrease on Paris (-0.0059). However, the most notable result is the synergistic effect of SV + QE on Paris. While QE alone is highly destructive to the ranking, the Spatial Verification effectively filters out the noise, turning a -0.0683 deficit into a $+0.0320$ gain over the baseline. Conversely, on Oxford, the baseline is already so saturated that even geometrically verified query expansion fails to outperform standard spatial re-ranking.

G. Global Comparison

Table XI consolidates the results of all five systems on both datasets.

The results confirm a clear performance hierarchy. Classical baselines achieve moderate mAP due to their inability to enforce geometric consistency between query and database

TABLE XI
GLOBAL COMPARISON – ALL SYSTEMS, BOTH DATASETS

Type	Method	Oxford5K	Paris6K
Classical	System 1	0.3634	0.4454
	System 2	0.4237	-
Proposed	System 3	0.4726	0.6168
	System 4	0.6680	0.7489
	System 5	0.7798	0.7587

images. The proposed state-of-the-art systems improve substantially, with spatial verification being the dominant contributor. The performance gap between Oxford5K and Paris6K is consistent across all methods, with Paris6K being systematically easier, likely due to less viewpoint variation and fewer confusing distractors.

V. DISCUSSION

A. System 1

1) *What Worked*: Spatial verification was the most important improvement: RANSAC promotes candidates with geometrically consistent matches, directly addressing false positives from repeated facades. Larger vocabularies ($K = 256 \rightarrow K = 4096$) reduced quantization ambiguity. Verified query expansion followed by a second spatial verification pass improved recall while controlling drift. Multi-scale SIFT gave a small but consistent final gain.

2) *What Did Not Work Strongly*: TF-IDF improved the baseline only slightly — weighting cannot enforce geometry. HOG/HSV fusion was marginal, as global descriptors fail under crop, viewpoint, and illumination changes. RootSIFT was useful but dataset-dependent, not a universal replacement for plain SIFT.

3) *Dataset Observations*: Paris consistently outperformed Oxford. Oxford benefited more from spatial verification, suggesting its false positives were geometrically inconsistent. Paris had a more reliable first-stage BoVW ranking.

4) *Failure Modes*: Repeated facades, occlusion, strong viewpoint change, ambiguous query crops, and similar architectural styles remain difficult even after multi-scale SIFT and repeated spatial verification. These limitations motivate richer methods such as ASMK [8], which preserve residual-level descriptor information beyond plain BoVW histogram counts.

B. System 2

The progression from HOG to Fisher Vector shows a large performance gap. The best HOG-only configuration obtained mAP 0.0350, while the best Fisher Vector configuration obtained mAP 0.4237. This indicates that Oxford Buildings is better handled by local descriptors aggregated through a probabilistic model than by a global gradient descriptor.

HOG is spatially rigid and depends strongly on global image layout. In landmark retrieval, the same building may appear under different viewpoints, crops, scales, and occlusions. This makes HOG less reliable as a standalone descriptor.

Fisher Vectors perform better because they aggregate many local RootSIFT descriptors. Local features are more robust to partial visibility and viewpoint changes. The GMM-based Fisher encoding also captures richer statistics than a simple global descriptor.

The fusion experiments show that HOG did not add useful information beyond Fisher in the tested settings. A small HOG weight produced almost the same result as Fisher alone, but did not surpass it. Larger HOG weights reduced retrieval quality. Therefore, the selected descriptor for this part of the project is the Fisher-only configuration with RootSIFT, PCA64, GMM $K = 64$, and improved Fisher normalization.

1) *Limitations:* The HOG parameter search was limited to a small set of image sizes, cell sizes, and orientation bins. The Fisher experiments tested only $K = 32$ and $K = 64$ with PCA fixed at 64 dimensions. A larger search could include different PCA dimensions, additional GMM sizes, and alternative fusion weights.

The fusion strategy is also simple. It uses weighted concatenation rather than learned weighting. A learned or validation-based fusion strategy could be explored.

Finally, this section evaluates descriptor quality using cosine similarity. It does not include query expansion or geometric re-ranking. These techniques could further improve retrieval, but they are outside the scope of this descriptor comparison.

C. System 3

Query expansion is the most consistent improvement across both datasets. By averaging the query representation with the top retrieved images, AQE incorporates complementary views of the same landmark that were absent from the original query crop. This is particularly valuable in Oxford and Paris, where the same building can be photographed from widely different angles and distances.

RANSAC spatial verification provides additional precision when the initial VLAD ranking already places relevant images in the top-50. However, if the initial ranking is weak or if relevant images share few keypoints with the query (due to occlusion, extreme viewpoint change, or repeated architectural patterns), RANSAC may demote true positives. The SIFT descriptor cache makes this stage feasible at query time without repeating the expensive extraction.

1) *Limitations:* The pipeline has several limitations. First, the vocabulary size $K = 256$ is small compared to large-scale systems that use $K > 100\,000$ visual words. A larger vocabulary would produce more discriminative centroids and tighter residuals, improving VLAD quality. Second, compressing 32 768-dimensional VLAD vectors to 512 dimensions retains only a fraction of the total variance; increasing PCA dimensionality or using a larger vocabulary would reduce this information loss. Third, SIFT and RANSAC are purely CPU-bound operations with no GPU acceleration path, making the pipeline slower than deep-learning alternatives. Fourth, the system is sensitive to repeated textures (*burstiness*): windows, arches, and brick patterns appear in many buildings and can produce false positive matches.

D. System 4

The experiments support three main observations. First, SIFT is the most effective local descriptor in this ASMK configuration. RootSIFT is competitive but does not surpass SIFT, which suggests that ASMK normalization and residual aggregation may reduce the advantage usually provided by the RootSIFT transformation. KAZE performs worse, indicating that preserving edges through nonlinear scale space does not necessarily translate into better ASMK retrieval under the selected parameters.

Second, query expansion is the most robust improvement. Its consistent positive effect indicates that the first ASMK ranking contains enough relevant images to enrich the query representation without severe drift. This is especially important in landmark retrieval, where different views of the same building may contribute complementary local evidence.

Third, spatial verification is not universally beneficial. Although geometric consistency is theoretically useful for eliminating false positives, the results show that RANSAC can reduce mAP when the initial ASMK ranking is already strong or when relevant images do not produce enough inliers. In Oxford, SV hurts performance in all evaluated combinations. In Paris, SV becomes useful only when combined with QE, suggesting that stronger candidate selection before verification improves the reliability of geometric re-ranking.

E. System 5

The evaluation of System 5 highlights three critical insights regarding advanced traditional retrieval pipelines. First, addressing burstiness is fundamental for architectural datasets. By combining Burst Filtering with RootSIFT and ASMK, the baseline mAP significantly outperforms all previous systems. Burst filtering effectively resolves the limitation noted in System 3, where repeated textures like windows and brickwork dominate the descriptor space, allowing the true geometric layout of the landmarks to drive the similarity score.

Second, the behavior of Query Expansion is entirely inverted compared to Systems 3 and 4. In earlier pipelines, the initial ranking was weak enough that average query expansion (AQE) consistently provided a robust improvement by pulling in complementary views. In System 5, the baseline ranking is already highly saturated and accurate. Consequently, blindly averaging the top retrieved images introduces severe query drift, as evidenced by the drop in mAP when QE is applied alone.

Third, while SV alone provides only marginal changes, the combined SV+QE configuration on Paris demonstrates synergy. By requiring geometric consistency before allowing an image to contribute to the expanded query, the system successfully filters out noise and extracts further performance gains.

1) *Limitations:* Despite its strong performance, this pipeline has notable limitations. The combination of high-resolution RootSIFT extraction, burst filtering, and large-vocabulary ASMK requires significant memory and storage overhead compared to the highly compressed VLAD or Fisher

Vector representations. Furthermore, the pipeline is highly sensitive to dataset-specific hyperparameter tuning; the strict SV+QE gating that succeeds on Paris fails to improve the saturated Oxford baseline. Finally, like the previous systems, this approach remains reliant on CPU-bound, handcrafted local features, lacking the end-to-end GPU acceleration and semantic robustness offered by modern deep learning descriptors.

VI. CONCLUSION

This work presents a comprehensive study of five handcrafted image retrieval pipelines on the Oxford5K and Paris6K Buildings benchmarks, covering two classical baselines and three progressively stronger local-feature systems. The experimental progression reveals a clear performance hierarchy driven by four key design decisions.

Spatial verification is the single most impactful component. Across all systems, RANSAC-based geometric re-ranking consistently produces the largest mAP gains over the initial ranking. BoVW and ASMK baselines retrieve images that share visual words with the query, but cannot enforce spatial consistency. Repeated facades, windows, and architectural patterns generate false positive matches that only geometric verification can suppress.

Descriptor quality sets the ceiling for all downstream components. The transition from plain SIFT to RootSIFT provides a consistent free improvement by replacing Euclidean distance with the Hellinger kernel. Fisher Vectors outperform HOG by a large margin (0.4237 vs. 0.0350 on Oxford5K), confirming that local descriptor aggregation is far more discriminative than global gradient structure for landmark retrieval. KAZE underperformed SIFT under the ASMK framework, suggesting that nonlinear scale-space edge preservation does not directly translate into better residual-aggregation retrieval.

Burst filtering unlocks the full potential of ASMK. System 5 demonstrates that early burst filtering through connected components, combined with RootSIFT and ASMK, raises the baseline from 0.6327 to 0.7701 mAP on Oxford5K before any post-processing. This represents the largest single architectural improvement in the project, confirming that burstiness—the tendency of repeated architectural elements to dominate the similarity score—is the primary obstacle in urban landmark retrieval.

Query expansion must follow, not precede, spatial verification. In Systems 1 and 3, plain query expansion before geometric re-ranking introduced query drift. In Systems 4 and 5, applying QE after spatially verified candidates consistently improved recall. System 5 shows the most striking example: unverified QE alone degrades mAP by -0.0388 on Oxford and -0.0683 on Paris, while SV+QE recovers and exceeds the baseline by $+0.0320$ on Paris.

The best overall results are achieved by System 5 (Burst Filtering + RootSIFT + ASMK), reaching **0.7798** mAP on Oxford5K with spatial verification and **0.7587** mAP on Paris6K with the full SV+QE pipeline. The performance gap between this system and the SIFT+BoVW baseline (0.3634 on Ox-

ford5K) quantifies the cumulative benefit of richer descriptor aggregation, burstiness handling, and geometric re-ranking.

Future work within the classical paradigm could explore soft and multi-assignment [8], joint PCA reduction over multiple vocabularies [4], and database-side feature augmentation [11] to further improve recall without sacrificing the geometric precision achieved by RANSAC verification.

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APPENDIX A QUALITATIVE EXAMPLES

A. System 1

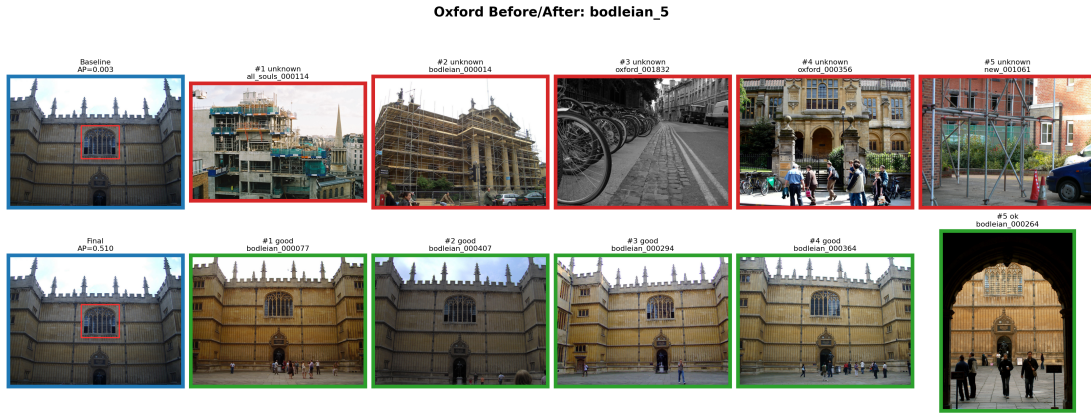


Fig. 8. Example of a good and bad retrieval using System 1

B. System 2



Fig. 9. Example of a good retrieval using System 2



Fig. 10. Example of a bad retrieval using System 2

C. System 3

pompidou_3 — top-5 retrievals (AP: 0.934)

Query

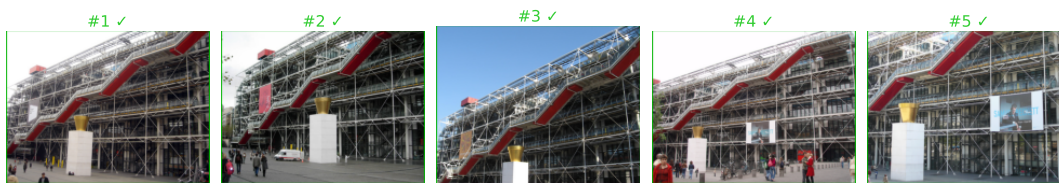


Fig. 11. Example of a good retrieval using System 3

magdalen_4 — top-5 retrievals (AP: 0.004)

Query

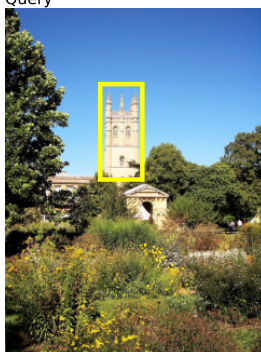


Fig. 12. Example of a bad retrieval using System 3

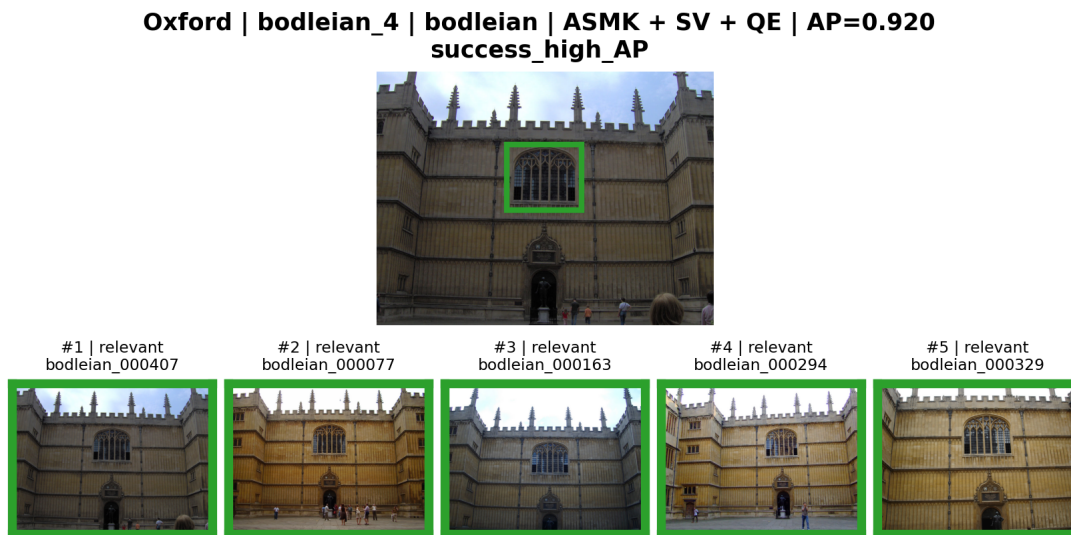


Fig. 13. Example of a good retrieval using System 4



Fig. 14. Example of a bad retrieval using System 4

ashmolean_2 — top-5 retrievals (AP: 0.777)



Fig. 15. Example of a good retrieval using System 5

magdalen_4 — top-5 retrievals (AP: 0.147)



Fig. 16. Example of a bad retrieval using System 5

APPENDIX B
PIPELINE DIAGRAMS

This appendix consolidates the workflow diagrams referenced in Section III.

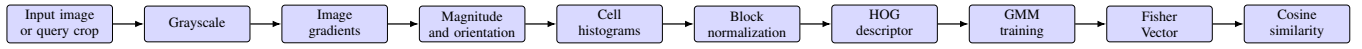


Fig. 17. Pipeline B1: HOG feature extraction, Fisher Vector encoding, and cosine-similarity retrieval. Queries use the official bounding-box crop and database images use the full frame.

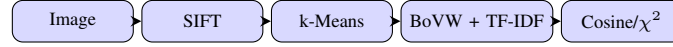


Fig. 18. Pipeline B2: SIFT + BoVW.



Fig. 19. RootSIFT + Fisher Vector pipeline for the Oxford5K descriptor-level study.

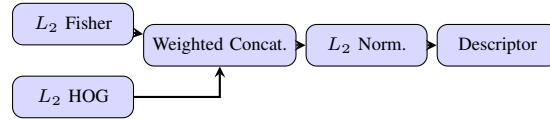


Fig. 20. HOG-Fisher fusion pipeline. The normalized Fisher and HOG vectors are weighted, concatenated, and normalized again before retrieval.

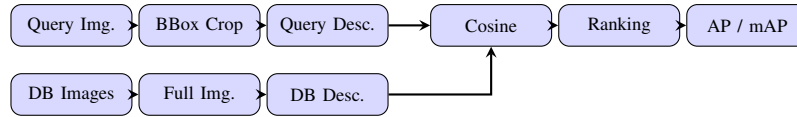


Fig. 21. Simplified retrieval protocol for the Oxford5K descriptor-level study. Queries use the official crop and database images use the full frame.

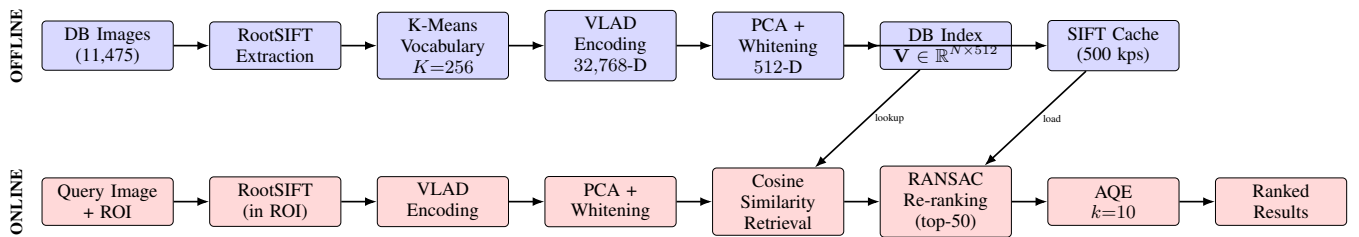


Fig. 22. Full retrieval pipeline. Blue: offline stage (run once, cached to disk). Red: online stage (run per query). The DB index and SIFT cache built offline feed directly into the online retrieval and RANSAC verification steps.

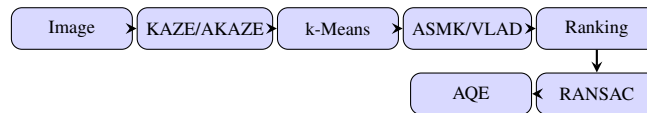


Fig. 23. Pipeline System 4: KAZE/AKAZE + ASMK/VLAD + RANSAC + AQE.